

Q) Consider a 10×10 input feature Map and a 3×3 filter with stride 2 and Zero-padding of size 1, followed by a max-pooling with a pooling window size of 2×2 and stride 2, Identify the dimensions of the output feature map?

Sol:

$$\text{Output size Formula} = \left[\frac{N - F + 2P}{S} \right] + 1$$

Where

$$N = 10 \text{ (Input size)}$$

$$F = 3 \text{ (Filter size)}$$

$$P = 1 \text{ (padding)}$$

$$S = 2 \text{ (stride)}$$

Step: 01 Convolution Output

$$= \left[\frac{(10 - 3 + 2(1))}{2} \right] + 1$$

$$= \left[\frac{9}{2} \right] + 1$$

$$= 4 + 1 = 5$$

So after the convolution, the feature map

5×5

Step: 02 Max-pooling Output

Pooling parameter

Pool size = 2×2

Stride = 2

Using the same formula (without padding)

$$= \left\lceil \frac{N-F}{S} \right\rceil + 1$$

$$= \left\lceil \frac{5-2}{2} \right\rceil + 1$$

$$= \left\lceil \frac{3}{2} \right\rceil + 1$$

$$= 1 + 1 = 2$$

So after max-pooling, the final output map size is

2×2

Ans

Convolution Learning:

①

3	3	2	1	0
0	0	1	3	1
3	1	2	2	3
2	0	0	2	2
2	0	0	0	1

5x5
Input

0	1	2
2	2	0
0	1	2

Kernel
3x3

- ① Kernel = Filter Detector
- ② stride
- ③ padding
- ④ pooling

Sol: ① Kernel

3	3	2	1	0
0	0	1	3	1
3	1	2	2	3
2	0	0	2	2
2	0	0	0	1

0	1	2
2	2	0
0	1	2

3x3

$$3 \times 0 + 3 \times 1 + 2 \times 2 + 0 \times 2 + 0 \times 2 + 1 \times 0 + 3 \times 0 + 1 \times 1 + 2 \times 2 =$$

$$0 + 3 + 4 + 0 + 0 + 0 + 0 + 1 + 4$$

12

Feature Map / convoluted Map

12	12	17
10	17	19
9	6	14

3x3

$$O = \lceil \frac{i-k}{2} \rceil + 1$$

$$= \lceil \frac{5-3}{2} \rceil + 1$$

$$= 2 + 1$$

$$= 3$$

Kernel Concept

② Stride

$$\begin{array}{|c|c|c|} \hline 12 & 12 & 7 \\ \hline 10 & 17 & 9 \\ \hline 9 & 6 & 14 \\ \hline \end{array}$$

$$O = \left\lfloor \frac{5-1}{5} \right\rfloor + 1$$

$$O = \left\lfloor \frac{5-3}{2} \right\rfloor + 1$$

$$O = \left\lfloor \frac{2}{2} \right\rfloor + 1$$

$$\begin{array}{|c|c|} \hline 12 & 7 \\ \hline 9 & 14 \\ \hline \end{array}$$

$$O = 2$$

proved

2x2

③ Padding

I can preserve the Information / Border Problem

In CNN, padding is a technique used to add extra pixels (usually zeros) around the border of an input image before apply a convolution.

			0	0	0
			2	1	0
			1	3	1
0	3	1	2	2	3
0	2	0	0	2	2
0	2	0	0	0	1
0	0	0	0	0	0

Input

0	1	2
2	2	0
0	1	2

Kernel

$$O = \left[\frac{i - k + 2p}{s} \right] + 1$$

$$\left[\frac{5 - 3 + 2 \times 1}{1} \right] + 1$$

$$O = \frac{5 - 3 + 2 + 1}{1}$$

$$O = 5 \times 5$$

④ Max-pooling (2x2)

Feature Map
after applying
padding
46/4 = 11.5

6	14	17	11	3
14	12	12	17	11
8	10	17	19	13
11	9	6	14	12
6	4	4	6	4

After pooling

14	17
11	19

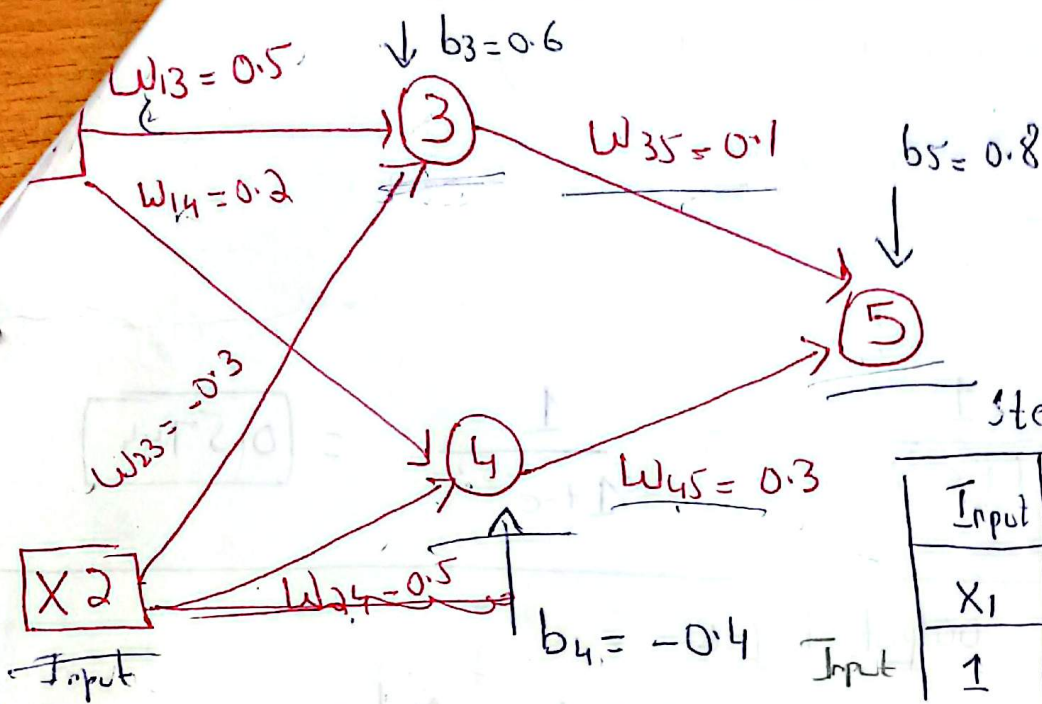
Max pooling

11.5	14.25
9.5	14.0

Average pooling

Back propagation numerical c-

learning
gate (1) = 0.5



Step (0)

Input	Output	Values
X_1	X_2	$T(\text{output})$
1	1	0 Target Output

(a) Calculate the Input (I) and Output (O) for each layer

For input layer	Output layer :-
$O_1 = I_1 = 1$	$\delta = \frac{1}{1 + e^{-x}}$
$O_2 = I_2 = 1$	

Sigmoid function

Step 1:- Initialization of Input and Output

Sol:-

For hidden layer

$$\begin{aligned}
 I_3 &= O_1 \times W_{13} + O_2 \times W_{23} + b_3 \\
 &= 1 \times 0.5 + 1 \times (-0.3) + 0.6 \\
 &= 0.5 - 0.3 + 0.6 \\
 &= 0.8 \checkmark
 \end{aligned}$$

$$\frac{\delta}{O_3} = \frac{1}{1 + e^{-I_3}} = \frac{1}{1 + e^{-0.8}} = 0.69 \checkmark$$

Step (3) In 11

$$I_4 = 0_1 \times W_{14} + 0_2 \times W_{24} + b_4$$

$$\begin{aligned} I_{net} &= 1 \times 0.2 + 1 \times 0.5 + (-0.4) \\ &= 0.2 + 0.5 - 0.4 \\ &= \boxed{0.3} \checkmark \end{aligned}$$

$$\begin{aligned} O_4 &= \frac{1}{1 + e^{-I_4}} = \frac{1}{1 + e^{-0.3}} = \boxed{0.5744} \\ \text{Output} \end{aligned}$$

For Output layer:-

$$\begin{aligned} I_5 &= \underline{0_3} \times \underline{W_{35}} + \underline{0_4} \times \underline{W_{45}} + b_5 \\ &= 0.69 \times 0.1 + 0.5744 \times 0.3 + 0.8 \\ &= 0.0690 + 0.1723 + 0.8 \\ &= \boxed{1.0413} \checkmark \end{aligned}$$

$$O_5 = \frac{1}{1 + e^{-I_5}} = \boxed{0.7391} \checkmark$$

Calculation of error at each node:

$$E_{ij} = O_j (1 - O_j) (T_j - O_j)$$

Output of each node Target = 0

For output layer:

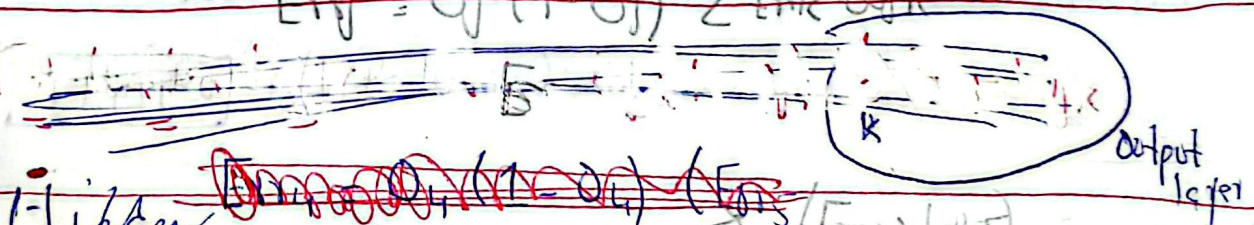
$$E_{15} = O_5 (1 - O_5) (T - O_5)$$

$$E_{15} = 0.7391 (1 - 0.7391) (0 - 0.7391)$$

$$E_{15} = 0.7391 (0.2609) (-0.7391)$$

$$E_{15} = \boxed{-0.1425}$$

$$E_{ij} = O_j (1 - O_j) \sum_k E_{ik} W_{jk}$$



layer $E_{14} = O_4 (1 - O_4) (E_{15} \times W_{45}) (0 - O_4)$

$$= 0.5744 (1 - 0.5744) (-0.1425 \times 0.3)$$

$$= 0.5744 (0.4256) (-0.04275)$$

$$= \boxed{-0.010451}$$

$$E_{13} = O_3 (1 - O_3) (E_{15} \times W_{35}) (0 - O_3)$$

$$= 0.69 (1 - 0.69) (-0.1425 \times 0.1)$$

$$= 0.69 (0.31) (-0.01425)$$

$$= \boxed{-0.0030481}$$

Step ③ Update the weights ^{beginning step}

$$W_{ij}(\text{new}) = W_{ij}(\text{old}) + I \times O_i \times E_{oj}$$

$$\begin{aligned} W_{45}(\text{new}) &= W_{45}(\text{old}) + I \times O_4 \times E_{45} \\ &= 0.3 + 0.5 \times 0.5744 \times (-0.1425) = \boxed{0.2591} \end{aligned}$$

$$\begin{aligned} W_{35}(\text{new}) &= W_{35}(\text{old}) + I \times O_3 \times E_{35} \\ &= 0.1 + 0.5 \times 0.69 \times (-0.1425) = \boxed{0.0508} \end{aligned}$$

$$\begin{aligned} W_{24}(\text{new}) &= W_{24}(\text{old}) + I \times O_2 \times E_{24} \\ &= 0.5 + 0.5 \times 1 \times (-0.010451) = \boxed{0.494775} \end{aligned}$$

$$\begin{aligned} W_{14}(\text{new}) &= W_{14}(\text{old}) + I \times O_1 \times E_{14} \\ &= 0.2 + 0.5 \times 1 \times (-0.010451) = \boxed{0.194774} \end{aligned}$$

$$W_{23}(\text{new}) = W_{23}(\text{old}) + I \times O_2 \times E_{23} = \boxed{-0.3015}$$

$$W_{13}(\text{new}) = W_{13}(\text{old}) + I \times O_1 \times E_{13} = \boxed{0.4984}$$

Step ④: Update the bias: $b_j = b_j(\text{old}) + I \times E_{nj}$

$$\begin{aligned} b_3(\text{new}) &= b_3(\text{old}) + I \times E_{13} \\ &= 0.6 + 0.5 \times (-0.030481) = \boxed{0.5984} \end{aligned}$$

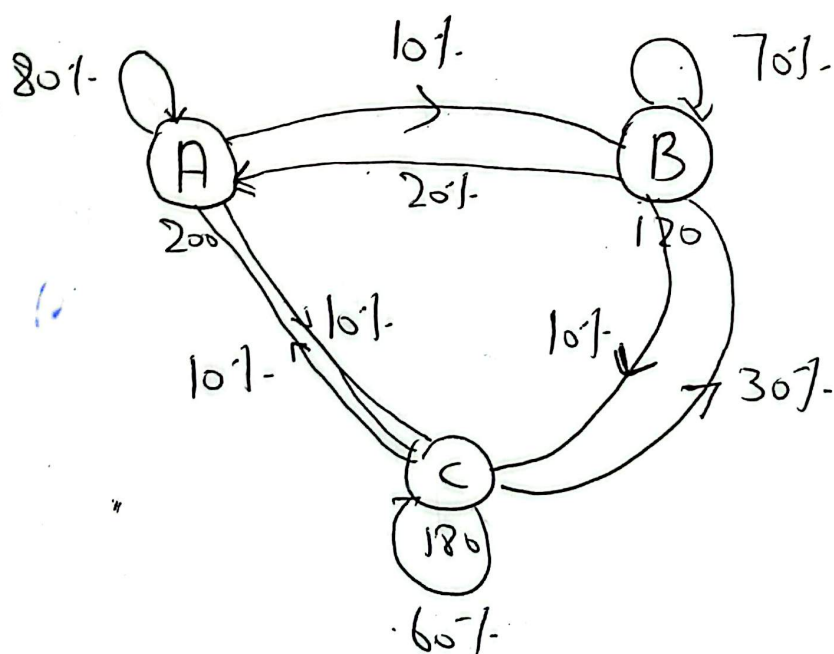
$$\begin{aligned} b_4(\text{new}) &= b_4(\text{old}) + I \times E_{14} \\ &= -0.4 + 0.5 \times (-0.010451) = \boxed{-0.4052} \end{aligned}$$

Markov Chain Numericals

Markov chain is a mathematical Model used in ML to predict what happen next based only on the current state, not the past history. A Markov Chain is a process where the next event depends only on the present event.

Example: If today weather is Raining, the chance of tomorrow being Raining or Sunny depend only on today weather, not on the weather from previous days.

Example: 01



$$X_i = P X_0$$

Sol:

$$[\text{NEXT STATE}] = \begin{bmatrix} \text{Matrix of} \\ \text{Transition} \\ \text{probabilities} \end{bmatrix} [\text{Current State}]$$

$$\begin{bmatrix} A_1 \\ B_1 \\ C_1 \end{bmatrix} = \begin{matrix} & \text{From} \\ & \begin{matrix} A & B & C \end{matrix} \\ \begin{bmatrix} A \\ B \\ C \end{bmatrix} & \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \end{matrix} \begin{matrix} \text{To} \\ \begin{matrix} AA & BA & CA \\ AB & BB & BC \\ CA & CB & CC \end{matrix} \end{matrix} \begin{bmatrix} A=200 \\ B=120 \\ C=180 \end{bmatrix} \begin{bmatrix} 0.40 \\ 0.24 \\ 0.36 \end{bmatrix}$$

$$\begin{bmatrix} A_1 \\ B_1 \\ C_1 \end{bmatrix} = \begin{matrix} & \text{From} \\ & \begin{matrix} A & B & C \end{matrix} \\ \begin{bmatrix} A \\ B \\ C \end{bmatrix} & \begin{bmatrix} 0.8 & 0.2 & 0.1 \\ 0.1 & 0.7 & 0.3 \\ 0.1 & 0.1 & 0.6 \end{bmatrix} \end{matrix} \begin{matrix} \text{To} \\ \begin{matrix} A \\ B \\ C \end{matrix} \end{matrix} \begin{bmatrix} A=200 \\ B=120 \\ C=180 \end{bmatrix} \begin{bmatrix} 0.40 \\ 0.24 \\ 0.36 \end{bmatrix} \downarrow$$

$$\begin{bmatrix} 0.404 \\ 0.316 \\ 0.298 \end{bmatrix} = \begin{bmatrix} (0.8)(0.4) + (0.2)(0.24) + (0.1)(0.36) \\ (0.1)(0.4) + (0.7)(0.24) + (0.3)(0.36) \\ (0.1)(0.4) + (0.1)(0.24) + (0.6)(0.36) \end{bmatrix}$$

~~1.018~~

1.000 ✓

500X Total

$$\begin{bmatrix} 202 \\ 158 \\ 140 \end{bmatrix} = \begin{bmatrix} 0.404 \\ 0.316 \\ 0.298 \end{bmatrix}$$

Ans

Means Clustering

Means Clustering for the following dataset for 2 clusters.

No	x_1	y_1
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77

$k=2$

	x_2	y_2
C1	185	72
C2	170	56

Sol:

$$D(x_n, y_n) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

No	x_1	y_1	C1 $x_2 = 185, y_2 = 72$	C2 $x_2 = 170, y_2 = 56$	Assignment
1	185	72	0	21.93	C1
2	170	56	21.93	0	C2
3	168	60	20.81	4.472	C2
4	179	68	7.211	15	C1
5	182	72	3	20	C1
6	188	77	5.83	27.66	C1

= Compute the new centroid for each
mean of both the cluster value (C1 & C2)

$$C1 = (185, 72), (179, 68), (182, 72), (188, 77)$$

$$\text{Mean}[x, y] = [(185 + 179 + 182 + 188)/4, (72 + 68 + 72 + 77)/4]$$

$$C1(\text{New}) = (\underset{x_2}{183.5}, \underset{y_2}{72.25})$$

$$C2 = (170, 56), (168, 60)$$

$$\text{Mean}[x, y] = [(170 + 168)/2, (56 + 60)/2]$$

$$C2(\text{New}) = (\underset{x_2}{169}, \underset{y_2}{58})$$

Step: 03

ND	x1	y2	C1	C2	Assignment
1	185	72	1.501	21.26	C1
2	170	56	21.13	2.24	C2
3	168	60	19.76	2.24	C2
4	179	68	6.2	14.142	C1
5	182	72	1.58	19.105	C1
6	188	77	6.54	26.9	C1

Ans 1